

ERWIN PAR-T

A Tree-based Hierarchical Transformer for Large Scale Physical Systems (Possibly for Jet Tagging)

05/21/26

Standard Self Attention Mechanism

$$Q, K, V = XW_q, XW_k, XW_v, \quad \text{Att}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{C'}} + B \right) V$$

- Particle Attention Block :

$$P - \text{MHA}(Q, K, V) = \text{SoftMax}(QK^T / \sqrt{d^K} + U)V$$

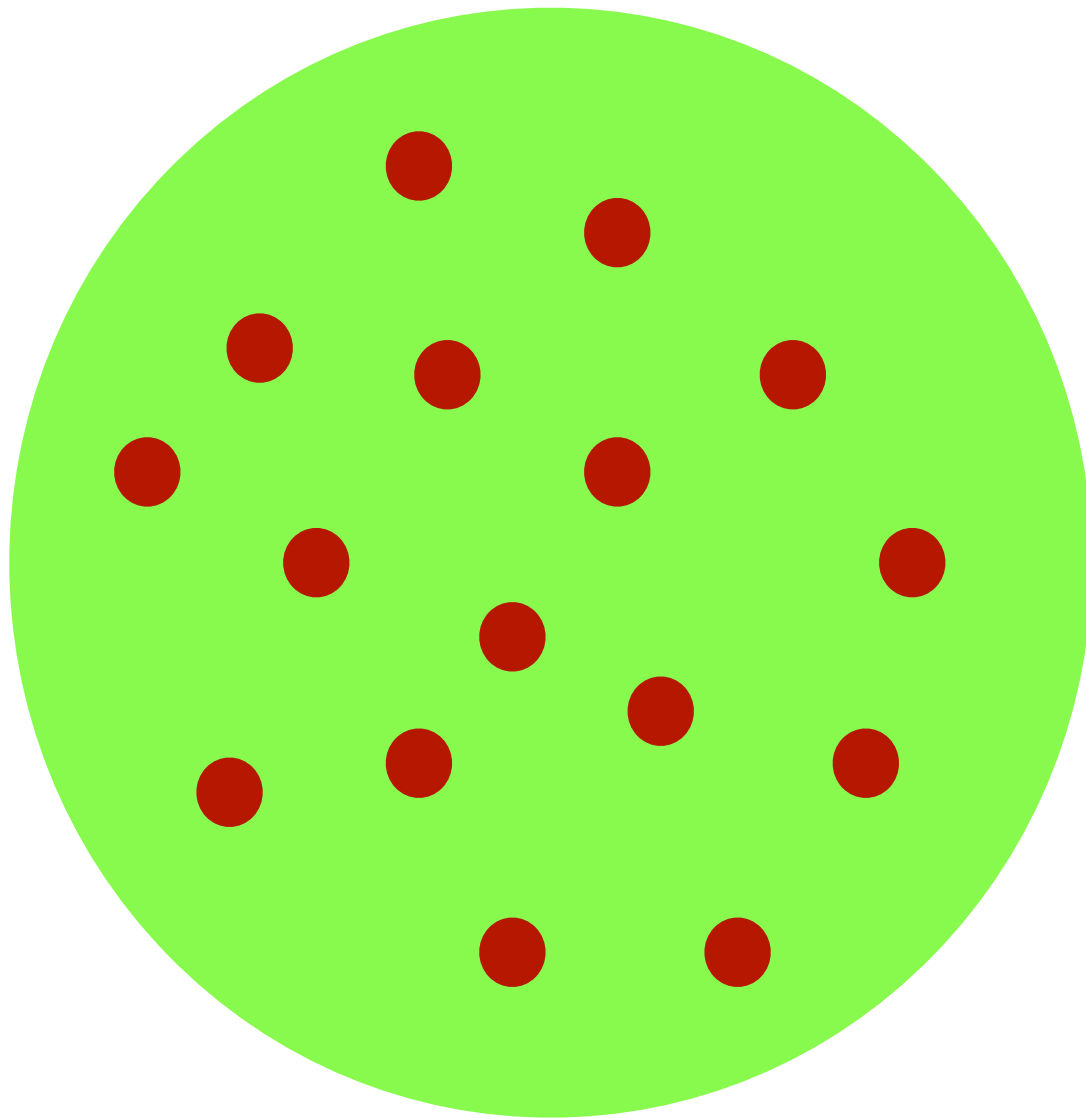
Where U is pre-softmax attention weight to incorporate particle attention feature.

- Without any locality constraints, computes interactions between all elements in the input set.
- In result, we get a quadratic computational cost $\mathcal{O}(N^2)$

Ball and Ball Tree in a Point Cloud

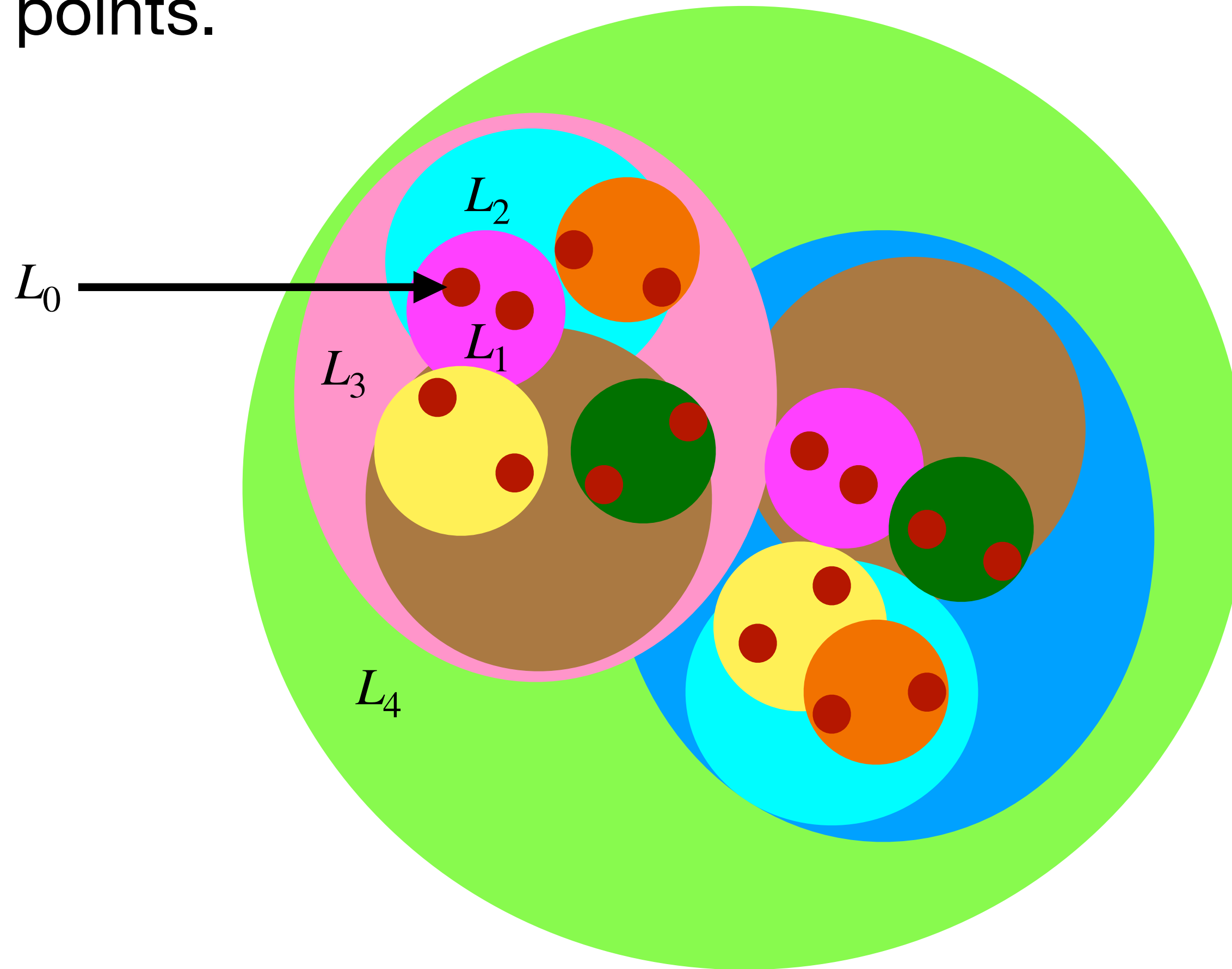
- A ball is a region bounded by a hypersphere in $\mathbb{R}^d$.
- Each ball is represented by the coordinates of its center $\mathbf{c}$ and radius r .

$$B = B(c, r) = \left\{ z \in \mathbb{R}^d \mid \|z - c\|_2 \leq r \right\}$$



Ball and Ball Tree in a Point Cloud

- A ball tree T on a point set P is a hierarchical sequence of partitions $\{L_0, L_1, \dots, L_m\}$.
- Each Level L_i cover disjoint balls that covers P , at leaf level ($i = 0$) the nodes are the original points.

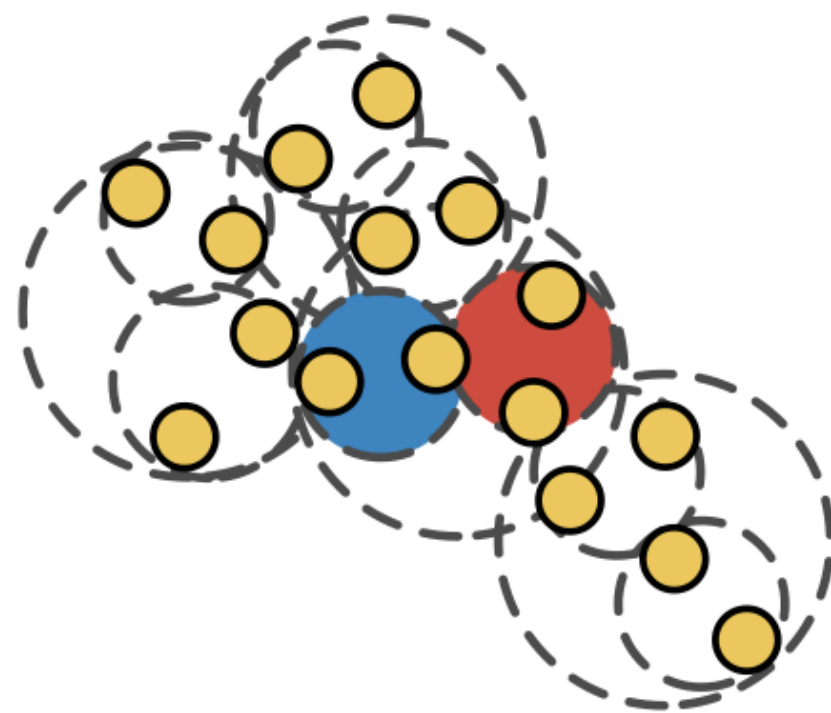


Ball Tree Attention

- For each ball attention operator we specify a level k of the ball tree where each ball $B \in L_k$ contains 2^k leaf nodes.
- That means larger balls will capture long range dependencies while smaller balls are more resources efficient.

$$leaves_B = \{B' \in L_{leaf}\}$$

Ball tree



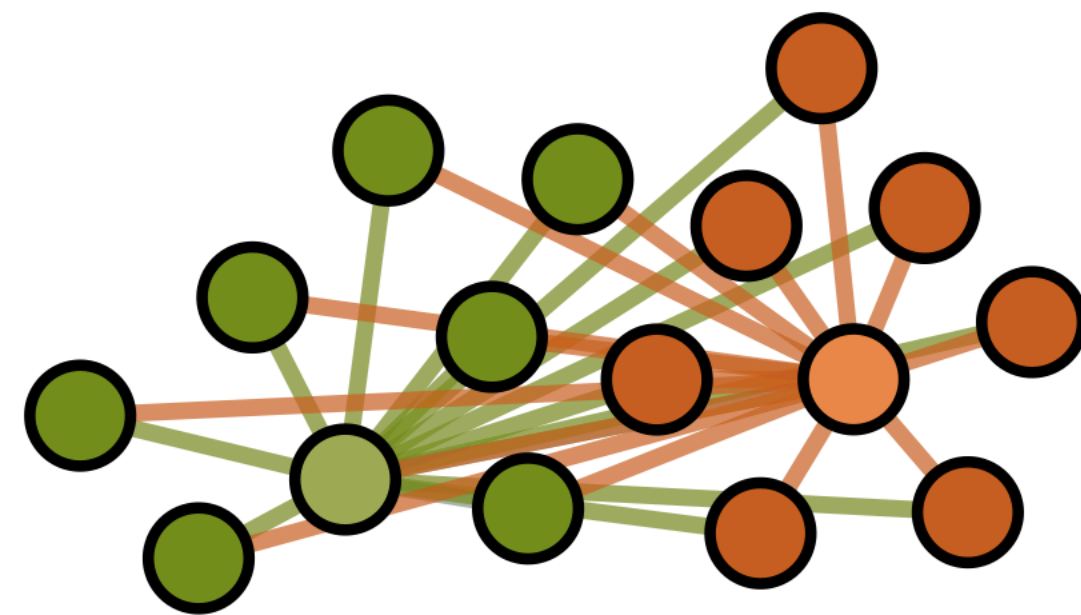
$$X_B = \{x'_B \in X_{leaf} \mid B' \in leaves_B\}$$

$$X'_B = BAtt(X_B) = Att(X_B W_q, X_B W_k, X_B W_v)$$

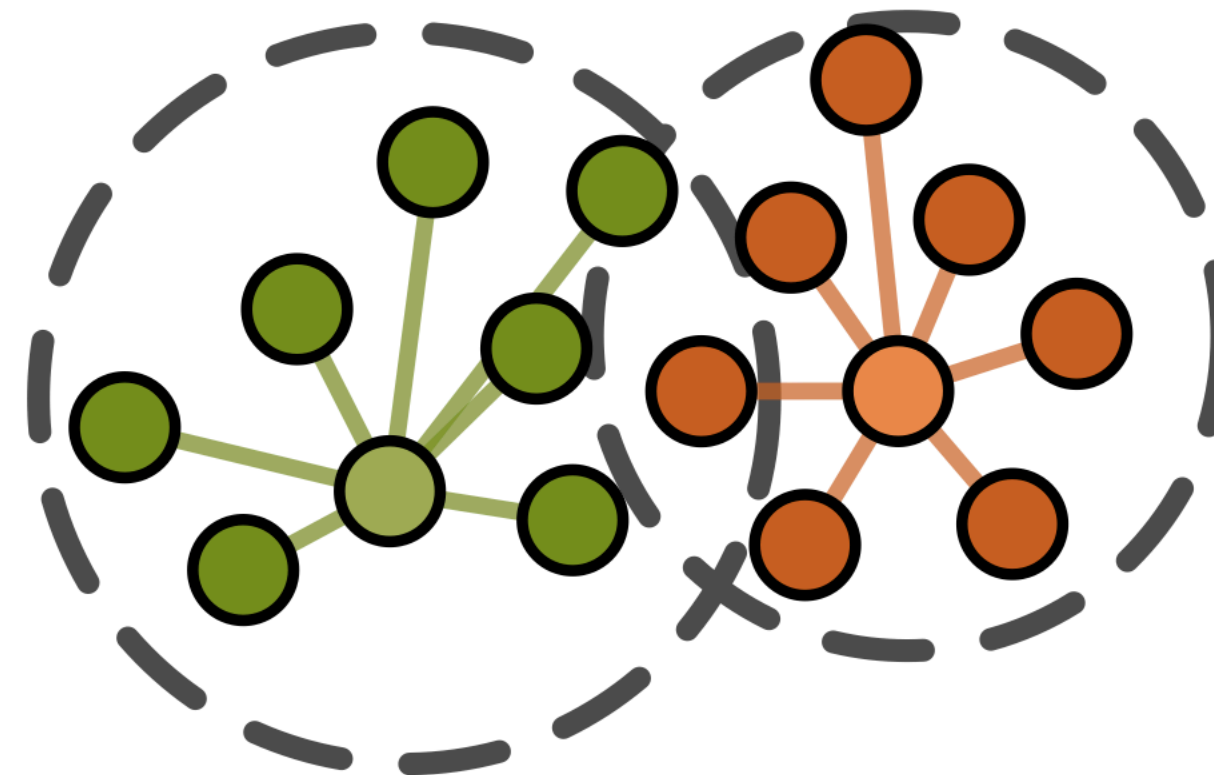
Computational Cost

- Now, as computation for attention is done independently for each ball. The computational cost reduced from quadratic to linear i.e $\mathcal{O}(B^2 \cdot \frac{n}{|B|})$.
Quadratic in ball size and linear in no of balls.

Attention $\mathcal{O}(n^2)$



Ball Attention $\mathcal{O}(n)$



ErwinParT with JetClass dataset.

- I made the Ball attention for our efficient particle transformer.
- Ran it with only 10% of JetClass dataset with RTX 4090

Model	Accuracy	AUC	H→bb ⁻ Rej 50%	H→cc ⁻ Rej 50%	H→gg Rej 50%	H→4q Rej 50%	H→ℓvqq' Rej 99%	t→bqq' Rej 50%	t→bℓv Rej 99.5%	W→qq' Rej 50%	Z→qq ⁻ Rej 50%
PFN	0.772	0.9714	2924	841	75	198	265	797	721	189	159
P-CNN	0.809	0.9789	4890	1276	88	474	947	2907	2304	241	204
ParticleNet	0.844	0.9849	7634	2475	104	954	3339	10526	11173	347	283
ParT (plain)	0.849	0.9859	9569	2911	112	1185	3868	17699	12987	384	311
ErwinParT	0.825	0.9819	8000	1681	97	590	1695	5556	4545	296	244
ParT	0.861	0.9877	10638	4149	123	1864	5479	32787	15873	543	402